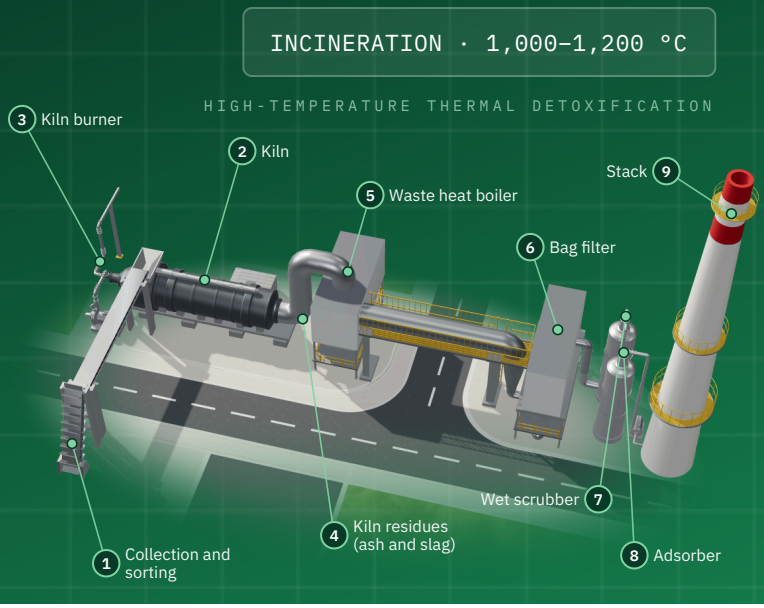
NATIONAL PROJECT

National Project of the Waste Management and Circular Economy Development Agency for the Establishment of a Specialized Facility for the Treatment and Disposal of Toxic Chemical Waste

A system for the detoxification, disposal and destruction of toxic chemical waste



01 · THE CURRENT PROBLEM

14
storage facilities

715 t
of obsolete pesticides in storage

35 thousand t
of soil exposed to contamination

7 billion m³
of water exposed to contamination

Key barriers. The existing disposal infrastructure is insufficient; the regulatory framework and the oversight system require comprehensive improvement.

Location	**Forish District, Jizzakh Region**
Technology	**Incineration, 1,000–1,200 °C**
Grant	**USD 5 million**
Investor selection	**Swiss challenge**

02 · INTERNATIONAL EXPERIENCE

Germany
Destruction of pesticide waste by incineration (POPs standard)

Japan
Incineration and high-temperature thermal technologies

France
Detoxification of hazardous waste in large incinerators

03 · WASTE MANAGEMENT PROCESS · SIMPLIFIED DIAGRAM

01 **Collection**
Separate collection of hazardous waste in special containers

02 **Transport**
Delivery by licensed special vehicles

03 **Processing**
Detoxification or recycling

04 **Disposal**
Detoxification through technological processes

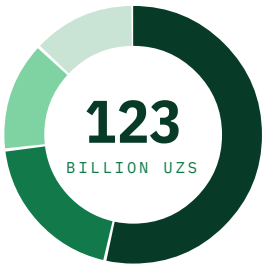
05 **Neutralisation**
Measures to neutralise residues

06 **Placement**
Placement of harmless residues at dedicated landfills

04 · PROJECT COSTS

5 million USD
From ILO grant funds

62 billion UZS
From Agency funds

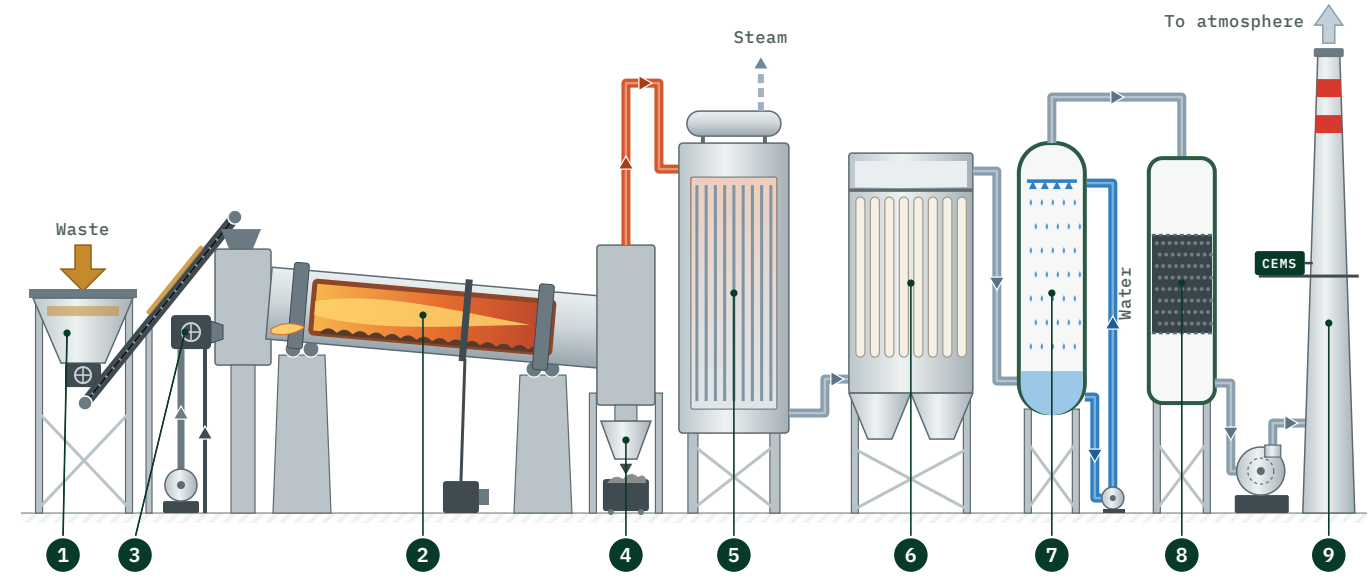


EQUIPMENT AND WORKS	BILLION UZS
Incineration kiln (rotary kiln)	66 54%
Chemical neutralisation and reagent preparation equipment	24 20%
Gas cleaning and filtration equipment	17 14%
Construction of buildings and structures on site	16 13%
Total	**123**

THE INSTALLATION · PROCESS FLOW DIAGRAM

Rotary kiln incineration. Multi-stage gas cleaning.

Toxic organic compounds are broken down at high temperature, and the resulting flue gases are cleaned in several stages. High-temperature incineration in a rotary drum kiln combined with comprehensive gas cleaning is one of the most versatile technologies for mixed industrial chemical waste.



Waste · Hot gas · Cooled gas · Water · Ash and slag · Steam

Section based on the process flow diagram · not to scale

1 Collection and sorting
Waste composition is determined (laboratory analysis); incompatible substances are kept in separate sealed areas; waste is fed to the kiln in metered doses

2 Kiln
Rotary drum type: an inclined steel drum with refractory lining · ~850–1,200 °C · solid waste 30–90 minutes

3 Kiln burner
Burners; the supply of waste, air and fuel is controlled automatically

4 Kiln residues (ash and slag)
Ash and slag are collected separately, tested in the laboratory and stabilised where necessary

5 Waste heat boiler
Heat is used to supply steam, hot water and process heat

6 Bag filter
Captures fly ash and dust

7 Wet scrubber
Removes acidic and soluble pollutants (HCl, HF, SO_2)

8 Adsorber
Activated carbon adsorbs mercury and organic micropollutants

9 Stack
Cleaned gas is released to the atmosphere; dust, CO, NO_x, SO_2 and HCl are measured continuously

PLANT PROCESS

From waste reception to placement of residues



Reception of toxic waste



Sorting and preparation



Neutralisation process



Incineration 1100–1200 °C



Gas cleaning and filtration



Safe placement of ash and residues

PLANT SPECIFICATIONS

Site area	**5.0 ha**
Building area	**30,000 m²**
Waste processing capacity	**20,000 – 30,000 tonnes**
Workforce	**120 people**
Operating mode	**Round the clock, 24/7**
Safety level	**In line with international standards**

ENVIRONMENTAL CONTROL AND SAFETY

Filtration system
cleaning of water, air and soil

Control
control of waste and hazardous substances

Minimal impact
minimal impact on the environment

Certificates
international safety certificates

SITE OF THE COMPLEX · COMPONENTS OF THE SYSTEM

Six functional zones and their purpose



1	Entry area: waste reception and weighing
2	Boiler house building (thermal treatment)
3	Chemical neutralisation and reagent processing building
4	Wastewater treatment and clean water facilities
5	Administration and utility building
6	Warehouse and secure storage area
A	Disinfection zone (washing complex)
B	Checkpoint



Entry area and disinfection corridor

- Mandatory washing and disinfection of vehicles before entry and exit
- Running gear and underbody washed with high-pressure water and disinfectant
- Inspection at the disinfection post on entry
- Biosafety is ensured and the spread of infection is prevented



Thermal destruction building

- High-temperature incineration (1100–1200 °C)
- Persistent organic pollutants (pesticides) are completely incinerated
- Two-stage gas cleaning system
- Ash and residues are detoxified separately



Chemical neutralisation and reagent preparation building

- Acid–alkali neutralisation process
- Reagent preparation and storage
- Automated dosing system
- Safe and sealed storage conditions



Wastewater treatment and storage facility

- Multi-stage biological and physico-chemical treatment
- The environmental impact of wastewater is minimised
- Treated water is reused for technical purposes



Administrative and auxiliary building

- Management and control centre
- Administrative and welfare facilities for staff
- Monitoring and reporting system



Warehouse and secure storage area

- Waste is stored safely until detoxification
- Warehouses with the highest level of protection

PROJECT IMPLEMENTATION WITH INVESTOR PARTICIPATION

Location
The complex will be established on the basis of the toxic chemical waste disposal landfill located in Egizbuloq mahalla, Forish District, Jizzakh Region.

Investor selection
Investors for the project will be selected through an international tender (Swiss challenge).

EXPECTED RESULTS

Through the annual disposal of

1,000 t

of waste:

50,000 t	**10 billion m³**	**150**	**1 billion UZS**
of soil protected from contamination	of water protected from pollution	permanent jobs created	annual budget revenue (taxes)

- Toxic chemical (pesticide) waste is completely destroyed
- Protection of the environment and human health is strengthened
- A sustainable and efficient system meeting international standards is created
- Economic efficiency and investment appeal increase
- A safe and sustainable environment is ensured